

Human Protein Atlas Image Classification Research

Sylvey Lin
yuhsinl2@illinois.edu

Zhi-Yi Cao
zihyic2@illinois.edu

1 Problem Definition

The Human Protein Atlas Image Classification project aims to enhance understanding and predictions of protein subcellular localization using multi-label classification methods. Protein localization within cells is vital to unraveling biological processes, understanding disease mechanisms, and identifying potential therapeutic targets. This Kaggle competition dataset provides fluorescence microscopy images where each image comprises multiple color channels representing distinct cellular structures, including the nucleus, microtubules, and the endoplasmic reticulum, along with the protein of interest.

Participants are tasked with accurately classifying the subcellular localization patterns of proteins from these complex immunofluorescence images. The multi-label nature of this challenge means that proteins may localize simultaneously in multiple cellular compartments, requiring advanced modeling strategies capable of identifying multiple concurrent labels. The competition uses the **macro F1 score** as the evaluation metric, which treats all classes equally regardless of frequency, making the task especially sensitive to class imbalance.

Achieving high accuracy in this task has significant implications, potentially accelerating research in areas such as drug discovery, disease diagnosis, and personalized medicine. However, the complexity of protein patterns, variable staining intensities, overlapping cellular structures, and inherent image noise pose significant hurdles. To overcome these, advanced image preprocessing techniques such as intensity normalization, contrast enhancement, and noise reduction are often necessary. Additionally, sophisticated deep learning methods, such as transformer-based architectures and specialized loss functions, are explored to handle the nuances and variability in the data.

2 Plan

Our approach builds upon existing high-performing solutions while investigating potential improvements using more recent architectures and loss functions. According to the leaderboard, the best solution was provided by the bestfitting team (Patil, 2019), who used a ResNet50 backbone followed by a classifier head trained with ArcFaceLoss (Deng et al., 2022). Although that approach was published in 2019, newer architectures and training strategies have emerged since then, prompting us to explore the following directions:

- **Backbone Architecture:** Compare ResNet-based models (such as ResNet18 and ResNet34 used in our baseline) with more recent architectures such as Swin Transformer (Liu et al., 2021) and Vision Transformer (ViT) (Dosovitskiy et al., 2021), to evaluate whether newer models offer performance advantages for HPA multi-label classification.
- **Loss Functions:** Our baseline models are trained using BCEWithLogitsLoss, a standard choice for multi-label classification. To explore potential improvements in inter-class separation and robustness, we will compare this with more advanced loss functions such as ArcFaceLoss (Deng et al., 2022), CurricularFace (Huang et al., 2020), and AdaFace (Kim et al., 2023), which have shown success in face recognition and fine-grained classification tasks.
- **Fine-tuning and Scheduling:** Explore different fine-tuning strategies and learning rate schedules (e.g., CosineAnnealingLR) to stabilize training and avoid overfitting, especially in low-frequency classes.

Through these investigations, we aim to develop a stronger baseline and identify the combination of

components that yield the best macro F1 score on this challenging dataset.

3 Key resources

In this project, we are using the following resources.

3.1 Dataset

The Human Protein Atlas (HPA) Image Classification dataset provides high-resolution microscopy images for classifying subcellular protein localization patterns. It was created using standardized confocal microscopy techniques and is widely used for studying protein functions and cellular structures.

The dataset contains 31,072 samples, each represented by four grayscale images corresponding to different fluorescence channels: nucleus (blue), microtubules (red), endoplasmic reticulum (yellow), and the protein of interest (green). The green channel is the primary signal for classification, while the other channels provide structural context. The images are available in 512x512 resolution for computational efficiency and high-resolution formats for more detailed analysis.

Each sample is labeled with one or more of 28 localization categories, making this a multi-label classification task. The dataset covers 27 different cell types, each with distinct morphology, influencing the observed protein patterns. A major challenge is class imbalance, requiring methods like weighted loss functions or augmentation to improve model performance.

This dataset serves as a benchmark for deep learning applications in biomedical imaging, particularly in classifying complex protein distributions. It supports the development of automated methods for analyzing cellular structures, with potential applications in computational biology and medical research.

3.2 Computing Platforms

We conduct our experiments on Google Colab Pro+, which provides access to high-performance GPUs (we mainly use A100) and ample memory resources. This platform supports our interactive development and enables efficient model training.

3.3 CodeBase

We implement our pipeline using PyTorch. Specifically, we use the built-in

`torchvision.models.resnet<version>` architecture with pre-trained weights on ImageNet as the backbone. The model is fine-tuned on the HPA dataset for multi-label classification. While we develop the overall framework and training loop from scratch, we rely on existing PyTorch utilities for model initialization, optimization, and evaluation.

We also incorporate custom components such as a multi-label classification head, a modified loss function (ArcFaceLoss), and additional pre-processing steps tailored to the four-channel input format of the HPA images. The ArcFaceLoss implementation is adapted with reference to the winning solution proposed by the bestfitting team in the Kaggle competition (Patil, 2019).

4 Evidence of Progress

We successfully ran pre-trained ResNet18 and ResNet34 models (from PyTorch's `torchvision.models` library) as our baseline models on the HPA dataset. Both models were fine-tuned on our multi-label classification task.

Hyperparameters. Our training setup included the following hyperparameters:

- Number of epochs: 30
- Initial learning rate: 0.001
- Learning rate decay: applied at epoch 15
- Batch size: 64

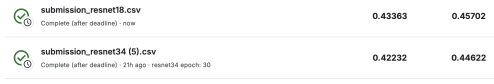
Model Setup. The baseline models were initialized with ImageNet-1K pre-trained weights and trained using the binary cross-entropy loss with logits (`nn.BCEWithLogitsLoss()`) as the criterion. We used the AdamW optimizer with a learning rate of 0.01.

Validation Strategy. To monitor training progress and model generalization, we split the training data into a 90%/10% train/validation set. Evaluation metrics were computed on this held-out validation set.

Kaggle Submission. We submitted our model predictions to the official Kaggle competition platform. As shown in Figure 1, both ResNet18 and ResNet34 achieved meaningful scores, confirming the correctness of our training and inference pipeline.

Model	Validation F1	Kaggle Public Score	Kaggle Private Score
ResNet18	0.6383	0.45702	0.43363
ResNet34	0.6258	0.44622	0.42232

Table 1: Model evaluation on held-out validation set and Kaggle test set.



submission_resnet18.csv Complete (after deadline) - now	0.43363	0.45702
submission_resnet34 (S).csv Complete (after deadline) - 21h ago - resnet34 epoch: 30	0.42232	0.44622

Figure 1: Kaggle submission results showing scores for ResNet34 and ResNet18 models.

5 Plan for the Remainder of the Project

Having completed the implementation and evaluation of our baseline models (ResNet18 and ResNet34 with BCEWithLogitsLoss), we now focus on improving performance through architectural and training refinements.

Our plan includes:

- **Model Comparison:** Evaluate newer transformer-based backbones such as Swin Transformer and ViT against our ResNet baselines using macro F1 score.
- **Loss Function Exploration:** Compare advanced loss functions—ArcFaceLoss, CurricularFace, and AdaFace—to understand their impact on label separation and imbalance handling.
- **Training Enhancements:** Apply learning rate schedules (e.g., CosineAnnealingLR) and tune hyperparameters to improve convergence.
- **Ablation and Final Submission:** Conduct ablation studies to identify key contributors to performance. Retrain the best model on the full dataset and submit final predictions to Kaggle.

Minimum goal: Achieve validation macro F1 comparable to baseline (around 0.45) with at least one alternative model/loss combination. **Maximum goal:** Outperform baseline on both validation and Kaggle test scores with improved architecture and loss function.

Risks and mitigation:

- *Risk:* Transformer models may require more compute and tuning. *Mitigation:* Start with smaller variants (e.g., Swin-Tiny) and limit experiment scope.

- *Risk:* New loss functions may not generalize well. *Mitigation:* Keep BCEWithLogitsLoss as a fallback and validate on multiple splits.

6 Member roles

As part of the project collaboration, the division of labor was as follows:

- **Zhi-Yi Cao** was responsible for writing the report
- **Sylvey Lin** focused on code implementation and searching for related work.

References

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